

PWA UX Benchmarking Report

Grocery List + Meal Planner

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A single-user, offline-first PWA combining a section-based grocery shopping list with live cart total and a weekly meal planner, used on phone in-store and on desktop for planning.

Method: 7-step evidence review against WCAG 2.1/2.2, Apple HIG, Material Design 3, Nielsen Norman Group, web.dev and MDN. Contrast ratios were computed directly from the current build's CSS tokens. Recommendations that conflict with the current build are flagged inline. Tier 3 community sources are labeled where used.

Section 1 - Executive Summary

Five changes move this app's experience the most, in priority order.

- 1. Fix two measured contrast failures.** Secondary data (price, weight, quantity) at #4a8462 measures 4.09:1, and component borders measure 1.4:1, both under WCAG AA. Lighten secondary text to #6fae86 and checkbox/input borders to #547e63.
- 2. Enlarge the highest-frequency targets.** The 30px checkbox and 20px delete control fall below Apple's 44pt and Material's 48dp. Expand the hit areas to 44px without changing the visuals.
- 3. Put repeated actions in the thumb zone.** Keep the pinned total. On phones, move the planner's top tabs to a bottom navigation bar.
- 4. Make destruction reversible.** Replace the hard-reset and item-delete confirms with an undo snapshot held in localStorage.
- 5. Raise small text to legible sizes:** 14px secondary data, 16px inputs (avoids iOS focus zoom), tabular numerals on every price.

Section 2 - Color and Visual Design

The current build ships a single dark theme (background #0e1910, surface #1a2b1e, primary text #dff2e3). Nielsen Norman Group's review of contrast-polarity studies found light mode produced better visual-acuity and proofreading performance across all ages, and that the light-mode advantage grew as font size dropped (NN/G, Dark Mode). For a list read at arm's length in a store, that favors offering a light theme rather than forcing dark. Dark mode still helps in dim conditions and for some low-vision users, while users with astigmatism can get halation on light-on-dark text (NN/G, Dark Mode Issues).

Recommendation: keep dark as the default if preferred, add a light theme, and honor the user's prefers-color-scheme. This is purely client-side and needs no backend.

Contrast minimums

WCAG 1.4.3 requires 4.5:1 for normal text and 3:1 for large text (18pt, or 14pt bold). WCAG 1.4.11 requires 3:1 for the visual boundaries of UI components and meaningful graphics (W3C). Store aisles have variable, often harsh lighting, so AA is a floor here, not a target. The table below reports ratios computed directly from the build's tokens.

	Token / use	Ratio	AA result	Recommended
	Primary text on bg	15.4:1	Pass	Keep
	Secondary text (price/qty/note)	4.09:1 bg / 3.39:1 card	FAIL (normal text)	Lighten to #6fae86 (5.73:1 on card)
	Borders / unchecked checkbox	1.69:1 / 1.40:1	FAIL (non-text 3:1)	#547e63 (3.22:1) for component edges
	Primary action / checked green	8.58:1	Pass	Keep
	Mode badge (amber)	6.98:1	Pass	Keep, keep text label

	Token / use	Ratio	AA result	Recommended
	Over-budget red	5.70:1	Pass	Keep, add a text label
	Check glyph / CTA text on green	2.10:1	FAIL	Use dark glyph #0e1910 (8.58:1)

• **Current build flag:** Secondary text #4a8462 (the price, weight, quantity and note text the user reads while shopping) fails AA at 4.09:1 on background and 3.39:1 on cards. This is the most-read data in the app.

• **Current build flag:** Borders and the unchecked checkbox outline at #2a4434 measure 1.4:1, far under the 3:1 needed for a component you must locate and tap.

• **Current build flag:** The white check glyph on the green fill is 2.1:1. It is saved only because the checked state is also shown by fill, strikethrough and dimming. Switch the glyph to dark for a clean 8.58:1.

Status colors

WCAG 1.4.1 (Level A) requires that color is never the only signal (W3C). The build mostly satisfies this: the checked state pairs green fill with a checkmark, strikethrough and dimmed text, and the mode toggle pairs amber with a text label. Two states need a non-color cue added: the over-budget indicator should add explicit text (for example OVER by \$4.20) rather than only turning the figure red, and the in-cart count should pair its color with a number badge.

Section 3 - Typography

Keep the existing two-family system: a system sans stack for labels and content, and a monospace stack for numbers. The build already does this (- - sans and - - mono), which is the right call: monospace digits are inherently tabular, so prices and totals align in a column and do not shift width as the total updates (MDN, font-variant-numeric). For any price rendered in the sans font, add font-variant-numeric: tabular-nums.

Material Design 3's type scale gives defensible minimums: body-large 16px, body-medium 14px, body-small 12px, label-medium 12px, label-small 11px (m3.material.io). 14px is a sensible floor for secondary data read at arm's length, and 16px is the floor for any text input because iOS Safari zooms the viewport when a focused input is under 16px (CSS-Tricks, Tier 3, documented iOS behavior).

Element	Recommended	Weight	Line height	Notes / current build
Item name (primary)	16px	600	1.3	Currently 15.5px. Bump to 16.
Secondary data (price, weight, unit)	14px	600	1.3	Currently 12.5px and below. Raise + tabular-nums.
Section labels (uppercase)	12px	700	1.2	Currently 12px. Keep.
Mode / tag badges	12px	700	1.2	Currently 10.5-11px. Raise to 12 (label-small 11 is the floor).
Text inputs (budget, price, add item)	16px	600	1.2	Add-item input is 15px: bump to 16 to stop iOS zoom.

Element	Recommend ed	Weight	Line height	Notes / current build
Running cart total	30px	800	1.0	Currently 30px mono. Keep.

Strikethrough state

Checked rows use strikethrough plus dimmed text. Strikethrough already reduces legibility, so the checked text must still clear 4.5:1, not drop to the failing #4a8462. Use the corrected #6fae86 for checked text and keep the line-through, the green fill and the checkmark as redundant cues (WCAG 1.4.3 and 1.4.1).

• **Current build flag:** The add-item input renders at 15px, which triggers iOS Safari's focus zoom. Raise to 16px.

Section 4 - Touch Targets and Mobile Interaction

Apple's Human Interface Guidelines set a 44x44pt minimum for interactive elements. Material Design sets 48x48dp (about 9mm physical), with at least 8dp between targets and the note that the tap area can extend past the visual element (Apple HIG; Material; Android Accessibility). WCAG 2.2 adds 2.5.8 Target Size (Minimum) at 24x24 CSS pixels for AA, while 2.5.5 (AAA) recommends 44x44 for primary touch actions (W3C). For an app whose core action is tapping rows while shopping, target 44px.

Element	Current	Standard	Result	Fix
Row checkbox (.check)	30x30px	44pt / 48dp	Below Apple & Material; passes WCAG 24px	Pad hit area to 44px; keep 30px visual
Delete control (.delbtn)	20x20px	44pt / 48dp	Fails all, incl. WCAG 24px	Raise to 44px, or move behind swipe + long-press
Mode toggle chip	~28px tall	44pt / 48dp	Below minimum	Pad to 44px height
Planner tabs	verify	44pt / 48dp	Check height	Set 44px min height

Thumb zone

Steven Hoober's 1,333-observation field study found 49% of people use a phone one-handed and 75% of interactions are thumb-driven, with the bottom-center the easy-reach zone and the top corners hard to reach (Hoober, via Smashing Magazine, Tier 2). Two consequences for this app: the pinned cart total sits correctly in the easy zone, and the planner's top tabs and the header's hide-completed and menu controls sit in the hard zone on a phone. Move repeated actions toward the bottom.

Swipe gestures

NN/G recommends limiting contextual swipe to destructive actions, and notes swipe has low discoverability and collides with the iOS back-swipe (NN/G, Contextual Swipe). Swipe-to-delete is acceptable for the item delete, but keep a visible delete control or a long-press alternative rather than hiding delete behind swipe alone.

One-handed accommodation

Because the user often holds a cart or basket, keep destructive actions (hard reset, delete) out of the resting-thumb path or guard them with undo, so an accidental thumb tap is cheap to recover (NN/G, Preventing User Errors).

Section 5 - Navigation and Layout Structure

NN/G's quantitative testing found hidden navigation roughly halves discoverability and raises task time versus visible navigation (NN/G, Hamburger Menus). The build's visible tabs are the right pattern: do not bury them in a hamburger.

The brief describes a 3-tab planner; the current build has 4 tabs (My Week, Recipes, Pantry, What's Available?). Four still fits Material's navigation bar, which supports 3 to 5 destinations with labels and badges and is intended for compact and medium widths (m3.material.io, Navigation bar).

Mobile (compact, under 600dp)

Move the planner's 4 tabs from the top to a bottom navigation bar so they land in the thumb zone, and keep the text labels visible rather than going icon-only (NN/G discoverability; Hoober). The shopping list is a single view and needs no tabs. The top-level switch between List and Plan (two separate pages today) should also be reachable at the bottom on a phone; with only 2 destinations a segmented control or a simple paired link is appropriate, since Material's bar wants 3 or more.

Footer pinning and cart total

Keep the cart total pinned to the bottom in large mono with the remaining/over figure. It is the most-glanced number and belongs in the easy zone, always visible. On non-installed mobile browsers the URL bar resizes the viewport, so rely on the existing fixed footer plus body bottom padding (the build pads 165px) and consider dvh units; the installed PWA avoids the URL-bar problem entirely.

Desktop breakpoint strategy

Material's window size classes are compact (under 600dp), medium (600 to 839dp) and expanded (840dp and up) (m3.material.io). At medium, a navigation rail replaces the bottom bar; at expanded (desktop, 1280px+), use a persistent sidebar or rail and show the planner's week and the recipe detail as two panes instead of a modal sheet. Cap content width so list rows do not stretch the full 1280px, which hurts scanning. Use `@media (pointer: coarse)` to keep 44px targets on touch and `(pointer: fine)` or `(any-hover: hover)` to allow denser rows and hover affordances on desktop (Smashing; CSS-Tricks; MDN).

- **Current build flag:** Planner tabs are top-anchored. On phones this puts primary navigation in the hard-to-reach zone; move to a bottom bar at compact widths.

Section 6 - Micro-interactions and Feedback

NN/G's response-time limits anchor every feedback decision: 0.1s feels instant, 1s keeps flow, 10s is the attention limit (NN/G, Response Times; Powers of 10). Because all data here is local, every interaction should resolve well under 0.1s with no spinner. Material's motion guidance puts small UI transitions around 150 to 200ms so a state change is perceptible but not slow (m3.material.io, Motion).

Interaction	Recommended pattern	Why / source
Row check / uncheck	Instant state change: fill box, strikethrough, dim, ~150ms ease; keep redundant cues.	Under 0.1s feels direct; color not the only signal (NN/G Response Times; WCAG 1.4.1).
Running total update	Update instantly on tap; optional 150-200ms count tween; tabular-nums so width is stable.	NN/G Response Times; Material Motion; MDN tabular-nums.
Mode toggle affordance	Show 3 modes as a segmented control with the active one visible, not a single cycling button.	Cycling hides the other states; a segmented control exposes them (iOS segmented control precedent; WCAG affordance).
Over-budget warning	Red figure plus explicit text 'OVER by \$X'; flip the remaining sign.	Color alone fails WCAG 1.4.1.
Empty states	Short guiding message plus the primary action (e.g. 'No items here yet. Add one.').	Empty state is a teachable first-use moment (Carbon Design System; NN/G).
Reset / delete	Offer undo (localStorage snapshot, ~6-10s toast) instead of a confirm dialog alone.	Undo beats confirmation for destructive actions (NN/G Confirmation Dialogs; User Mistakes).

The empty state matters more than it looks: NN/G notes blank screens appear constantly through filtering, clearing and resetting. Hide-completed, hard-reset and an empty section all reach zero in this app, so each needs a one-line prompt rather than a bare gap.

Section 7 - PWA-Specific UX

Install prompt

Do not fire the prompt on load. Capture `beforeinstallprompt`, defer it, and surface a custom Add-to-home affordance after an engagement signal such as a repeat visit or after the user generates a list (web.dev, Installation Prompt; Customize Install). iOS Safari does not fire `beforeinstallprompt`, so iOS needs manual 'Share, then Add to Home Screen' guidance instead.

iOS safe area

The build already sets `viewport-fit=cover` and uses `env(safe-area-inset-top/bottom)` in the header and footer, which is correct: without `viewport-fit=cover` the insets resolve to 0 and Safari letterboxes the app. The bottom home-indicator inset is about 34px on notched iPhones, and the footer already pads it (MDN, `env()`; iOS PWA notes, Tier 3). Verify on a physical device, since simulators report insets

unreliably.

Loading and theme color

A standalone launch shows a splash built from the manifest name, icon and background_color. The manifest's background_color (#15201b) differs from the app background (#0e1910), which produces a color flash on launch. Match them. Render the cached shell immediately (the service worker is already network-first with cache fallback) and avoid a spinner, since data is local.

Offline communication

The app is localStorage plus a network-first service worker with cache fallback, so it already works offline. With no backend, offline is the normal state and needs almost no messaging. Avoid alarming offline errors; at most show a quiet 'showing your saved list' note (web.dev, App Design).

Back navigation (iOS vs Android differ)

In standalone mode iOS shows no system back button, while Android always provides a system back (firt.dev; web.dev). Two consequences: every screen, including the recipe detail sheet, needs a visible in-app back or close control for iOS; and on Android you should push a history entry when opening the recipe sheet and handle popstate, so the system back closes the sheet instead of exiting the app.

• **Current build flag:** Manifest background_color #15201b does not match the app background #0e1910, causing a launch flash. The page theme-color (#080d09) also differs from the manifest theme_color (#15201b); align all three.

Section 8 - Source Log

Tier 1 = WCAG, Apple HIG, Material Design 3, Nielsen Norman Group, Baymard. Tier 2 = web.dev, MDN, Smashing. Tier 3 (community, documented behavior) labeled. Accessed June 12, 2026.

Section 2 - Color and Visual Design

- W3C - WCAG 2.1, SC 1.4.3 Contrast (Minimum) - <https://www.w3.org/TR/WCAG21/>
- W3C - WCAG, SC 1.4.11 Non-text Contrast - <https://www.w3.org/WAI/WCAG21/Understanding/non-text-contrast.html>
- W3C - WCAG, SC 1.4.1 Use of Color - <https://www.w3.org/WAI/WCAG21/Understanding/use-of-color.html>
- WebAIM - Contrast and Color Accessibility - <https://webaim.org/articles/contrast/>
- NN/G - Dark Mode vs. Light Mode: Which Is Better? - <https://www.nngroup.com/articles/dark-mode/>
- NN/G - Dark Mode: How Users Think About It and Issues to Avoid - <https://www.nngroup.com/articles/dark-mode-users-issues/>

Section 3 - Typography

- Material Design 3 - Typography / Applying type - <https://m3.material.io/styles/typography/applying-type>
- MDN - font-variant-numeric (tabular-nums) - <https://developer.mozilla.org/en-US/docs/Web/CSS/font-variant-numeric>
- CSS-Tricks - 16px or Larger Text Prevents iOS Form Zoom (Tier 3) - <https://css-tricks.com/16px-or-larger-text-prevents-ios-form-zoom/>
- W3C - WCAG 2.1 SC 1.4.3 (strikethrough contrast) - <https://www.w3.org/TR/WCAG21/>

Section 4 - Touch Targets and Mobile Interaction

- Apple - Human Interface Guidelines, Accessibility (44pt targets) - <https://developer.apple.com/design/human-interface-guidelines/accessibility>
- Material Design - Accessibility, touch targets 48dp - <https://m2.material.io/design/usability/accessibility.html>
- Google - Android Accessibility, touch target size - <https://support.google.com/accessibility/android/answer/7101858>
- W3C - WCAG 2.2 SC 2.5.8 Target Size (Minimum), 24px - <https://www.w3.org/WAI/WCAG22/Understanding/target-size-minimum.html>
- W3C - WCAG 2.1 SC 2.5.5 Target Size (Enhanced), 44px - <https://www.w3.org/WAI/WCAG21/Understanding/target-size.html>

- Smashing Magazine - The Thumb Zone (Hoover field study), Tier 2 - <https://www.smashingmagazine.com/2016/09/the-thumb-zone-designing-for-mobile-users/>
- NN/G - Using Swipe to Trigger Contextual Actions - <https://www.nngroup.com/articles/contextual-swipe/>
- NN/G - Preventing User Errors: Avoiding Conscious Mistakes - <https://www.nngroup.com/articles/user-mistakes/>

Section 5 - Navigation and Layout Structure

- NN/G - Hamburger Menus and Hidden Navigation Hurt UX Metrics - <https://www.nngroup.com/articles/hamburger-menus/>
- NN/G - Basic Patterns for Mobile Navigation - <https://www.nngroup.com/articles/mobile-navigation-patterns/>
- Material Design 3 - Navigation bar (3-5 destinations, badges) - <https://m3.material.io/components/navigation-bar/overview>
- Material Design 3 - Navigation rail / window size classes - <https://m3.material.io/components/navigation-rail/guidelines>
- Smashing Magazine - A Guide To Hover And Pointer Media Queries - <https://www.smashingmagazine.com/2022/03/guide-hover-pointer-media-queries/>
- MDN - @media pointer / any-hover - <https://developer.mozilla.org/en-US/docs/Web/CSS/@media/pointer>

Section 6 - Micro-interactions and Feedback

- NN/G - Response Time Limits: 0.1s, 1s, 10s - <https://www.nngroup.com/articles/response-times-3-important-limits/>
- NN/G - Powers of 10: Time Scales in UX - <https://www.nngroup.com/articles/powers-of-10-time-scales-in-ux/>
- Material Design 3 - Motion, easing and duration - <https://m3.material.io/styles/motion/easing-and-duration/tokens-specs>
- NN/G - Confirmation Dialogs Can Prevent User Errors (If Not Overused) - <https://www.nngroup.com/articles/confirmation-dialog/>
- Carbon Design System - Empty states pattern - <https://carbondesignsystem.com/patterns/empty-states-pattern/>

Section 7 - PWA-Specific UX

- web.dev - Installation prompt - <https://web.dev/learn/pwa/installation-prompt>
- web.dev - Provide your own in-app install experience - <https://web.dev/articles/customize-install>
- web.dev - App design - <https://web.dev/learn/pwa/app-design>
- MDN - Making PWAs installable - https://developer.mozilla.org/en-US/docs/Web/Progressive_web_apps/Guides/Making_PWAs_installable
- MDN - env() safe-area-inset - <https://developer.mozilla.org/en-US/docs/Web/CSS/env>
- firt.dev (M. Firtman) - PWA design tips: iOS vs Android back, Tier 3 - <https://firt.dev/pwa-design-tips/>
- ITNEXT/dev.to - Make Your PWAs Look Handsome on iOS (34px home indicator), Tier 3 - <https://itnext.io/make-your-pwas-look-handsome-on-ios-fd8dfcd5777>